

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: CSA51

COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO3: Examine cryptographic hash algorithms, digital signature schemes, and assess Kerberos and X.509 authentication frameworks for ensuring integrity, authentication, and non-repudiation.CO (BL4 , BL5)

Assessment Tool Weightage: 10%

QUESTIONS: 50

TOTAL MARKS: 50

Annexure A

MCQ

Code of Conduct:

I _Ch.Naga Sai Srikanth(192311159) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Ch.nagasaisrikanth

Signature of the student:

Annexure A

MCQ

1. A system uses MD5 for integrity checking. Which attack can most likely compromise it?

- A. Replay attack
- B. Collision attack
- C. Phishing attack

D. Brute force on keys

2. Which modification improves security when replacing MD5 in a password system?

- A. Use plaintext storage
- B. Use SHA-256 with salting
- C. Use shorter hash values
- D. Disable hashing

3. In a digital signature system, what ensures non-repudiation?

- A. Symmetric key
- B. Private key of sender
- C. Public key of receiver
- D. Hash length

4. Which scenario indicates a failure of a hash function?

- A. Slow computation
- B. Same hash for different inputs
- C. Large output size
- D. Deterministic output

5. HMAC is preferred over simple hashing because it:

- A. Reduces computation time
- B. Uses public keys
- C. Combines key with hash
- D. Eliminates encryption

6. In Kerberos, what prevents replay attacks?

- A. Password reuse
- B. Timestamp mechanism
- C. Hash collisions
- D. Key length

7. Which component in Kerberos issues tickets?

- A. Client
- B. Server
- C. KDC
- D. Firewall

8. X.509 certificates primarily provide:

- A. Data compression
- B. Authentication
- C. Routing
- D. Encryption only

9. What is the main weakness of MD5?

- A. Slow hashing
- B. Collision vulnerability
- C. Large key size
- D. Complex implementation

10. SHA-256 is preferred over MD5 because it:

- A. Is faster
- B. Has shorter output
- C. Is collision resistant
- D. Uses symmetric keys

11. In digital signatures, hashing is used to:

- A. Encrypt message
- B. Reduce message size
- C. Generate keys
- D. Authenticate network

12. Which attack targets MAC without key knowledge?

- A. Collision attack
- B. Dictionary attack
- C. Birthday attack
- D. Phishing

13. What is required for verifying a digital signature?

- A. Sender's private key
- B. Receiver's private key
- C. Sender's public key
- D. Shared secret key

14. In ElGamal signature, security depends on:

- A. Hash length
- B. Discrete logarithm problem
- C. RSA problem
- D. AES encryption

15. Schnorr protocol is mainly used for:

- A. Encryption
- B. Authentication
- C. Compression
- D. Routing

16. Which property ensures hash uniqueness?

- A. Determinism
- B. Collision resistance
- C. Speed
- D. Length

17. A weak MAC allows:

- A. Faster encryption
- B. Message tampering
- C. Reduced bandwidth
- D. Secure authentication

18. Kerberos relies on which cryptographic method?

- A. Asymmetric only
- B. Symmetric key cryptography
- C. Hashing only
- D. Quantum cryptography

19. X.509 certificates include:

- A. Private key
- B. Public key
- C. Session key
- D. Hash key

20. Which attack exploits hash collisions?

- A. MITM
- B. Birthday attack
- C. Replay attack
- D. DoS

21. Digital Signature Standard (DSS) is based on:

- A. AES
- B. DSA
- C. DES
- D. RC4

22. Which ensures integrity in HMAC?

- A. Encryption
- B. Keyed hashing
- C. Compression
- D. Encoding

23. In Kerberos, TGT stands for:

- A. Ticket Granting Ticket
- B. Trusted Gateway Token
- C. Temporary Grant Ticket
- D. Token Generation Table

24. Which protocol uses certificates extensively?

- A. Kerberos
- B. X.509
- C. HMAC
- D. MD5

25. Which property is violated if attacker finds two inputs with same hash?

- A. Pre-image resistance
- B. Collision resistance
- C. Determinism
- D. Efficiency

26. ElGamal signatures are vulnerable if:

- A. Key reused
- B. Random value reused
- C. Hash used
- D. Public key shared

27. Which improves MAC security?

- A. Removing key
- B. Increasing key randomness
- C. Using plaintext
- D. Reducing hash size

28. Which is NOT a hash property?

- A. Deterministic
- B. Fixed length
- C. Reversible
- D. Fast computation

29. Kerberos authentication fails if:

- A. Time not synchronized
- B. Hash too long
- C. Key too short
- D. Network fast

30. X.509 trust depends on:

- A. Hash length
- B. Certificate authority
- C. Encryption speed
- D. MAC value

31. Schnorr signatures are considered efficient because:

- A. Smaller signature size
- B. Larger keys
- C. No hashing
- D. No authentication

32. Which is a MAC algorithm?

- A. SHA-256
- B. HMAC
- C. RSA
- D. ElGamal

33. Digital signatures provide:

- A. Confidentiality only
- B. Integrity and authentication
- C. Compression
- D. Routing

34. Which algorithm is obsolete due to security flaws?

- A. SHA-256
- B. MD5
- C. HMAC
- D. DSS

35. In Kerberos, session keys are used for:

- A. Permanent storage
- B. Temporary communication
- C. Hashing
- D. Compression

36. Which ensures sender authenticity in email?

- A. Hash only
- B. Digital signature
- C. Encryption only
- D. MAC only

37. Which attack is prevented by HMAC?

- A. Replay attack
- B. Message tampering
- C. DoS
- D. Phishing

38. X.509 certificates are validated using:

- A. Symmetric keys
- B. Public key infrastructure
- C. Hash only
- D. MAC

39. Which is true about SHA algorithms?

- A. Reversible
- B. Collision resistant
- C. Symmetric
- D. Encrypted

40. Which parameter is critical in ElGamal security?

- A. Prime number
- B. Hash length
- C. MAC key
- D. Timestamp

41. Which step verifies a digital signature?

- A. Encrypt hash
- B. Decrypt signature with public key
- C. Generate MAC
- D. Compress message

42. Weak hash functions affect:

- A. Bandwidth
- B. Integrity
- C. Routing
- D. Compression

43. Kerberos uses which server?

- A. DNS
- B. KDC
- C. HTTP
- D. FTP

44. Which provides integrity and authenticity together?

- A. Encryption
- B. MAC
- C. Routing
- D. Compression

45. Schnorr protocol relies on:

- A. RSA problem
- B. Discrete log problem
- C. AES
- D. DES

46. Which is vulnerable to collision attacks?

- A. MD5
- B. SHA-512
- C. HMAC
- D. DSS

47. Which improves digital signature security?

- A. Larger key size
- B. Smaller hash
- C. No hashing
- D. Plaintext

48. Which ensures certificate authenticity?

- A. MAC
- B. CA signature
- C. Hash only
- D. Encryption

49. What is the main role of MAC?

- A. Confidentiality
- B. Authentication
- C. Compression
- D. Routing

50. Which combination is most secure for integrity?

- A. MD5 only
- B. SHA + HMAC
- C. Plaintext
- D. Encryption only

Answers

1. A system uses MD5 for integrity checking. Which attack can most likely compromise it?

- A. Replay attack
- B. Collision attack
- C. Phishing attack
- D. Brute force on keys

Answer: B. Collision attack

2.Which modification improves security when replacing MD5 in a password system?

- A. Use plaintext storage
- B. Use SHA-256 with salting
- C. Use shorter hash values
- D. Disable hashing

Answer: B. Use SHA-256 with salting

3.In a digital signature system, what ensures non-repudiation?

- A. Symmetric key
- B. Private key of sender
- C. Public key of receiver
- D. Hash length

Answer: B. Private key of sender

4.Which scenario indicates a failure of a hash function?

- A. Slow computation
- B. Same hash for different inputs
- C. Large output size
- D. Deterministic output

Answer: B. Same hash for different inputs

5.HMAC is preferred over simple hashing because it:

- A. Reduces computation time
- B. Uses public keys
- C. Combines key with hash
- D. Eliminates encryption

Answer: C. Combines key with hash

6.In Kerberos, what prevents replay attacks?

- A. Password reuse
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- C. Hash collisions
- D. Key length

Answer: B. Timestamp mechanism

7.Which component in Kerberos issues tickets?

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- D. Firewall

Answer: C. KDC

8.X.509 certificates primarily provide:

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- B. Authentication
- C. Routing
- D. Encryption only

Answer: B. Authentication

9.What is the main weakness of MD5?

- A. Slow hashing
- B. Collision vulnerability
- C. Large key size
- D. Complex implementation

Answer: B. Collision vulnerability

10.SHA-256 is preferred over MD5 because it:

- A. Is faster
- B. Has shorter output
- C. Is collision resistant
- D. Uses symmetric keys

Answer: C. Is collision resistant

11.In digital signatures, hashing is used to:

- A. Encrypt message
- B. Reduce message size
- C. Generate keys
- D. Authenticate network

Answer: B. Reduce message size

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Answer: C. Birthday attack

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Answer: C. Sender's public key

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- C. RSA problem
- D. AES encryption

Answer: B. Discrete logarithm problem

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- C. Compression
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Answer: B. Authentication

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- C. Speed
- D. Length

Answer: B. Collision resistance

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Answer: B. Message tampering

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- D. Hash key

Answer: B. Public key

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Answer: B. Birthday attack

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Answer: B. DSA

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Answer: B. Keyed hashing

23.In Kerberos, TGT stands for:

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Answer: A. Ticket Granting Ticket

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Answer: B. X.509

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Answer: A. Time not synchronized

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- C. Encryption speed
- D. MAC value

Answer: B. Certificate authority

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- B. Larger keys
- C. No hashing
- D. No authentication

Answer: A. Smaller signature size

32.Which is a MAC algorithm?

- A. SHA-256
- B. HMAC
- C. RSA
- D. ElGamal

Answer: B. HMAC

33.Digital signatures provide:

- A. Confidentiality only
- B. Integrity and authentication
- C. Compression
- D. Routing

Answer: B. Integrity and authentication

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- C. HMAC
- D. DSS

Answer: B. MD5

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- C. Hashing
- D. Compression

Answer: B. Temporary communication

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- C. Encryption only
- D. MAC only

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Answer: B. Decrypt signature with public key

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Answer: B. Integrity

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Answer: B. MAC

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- B. SHA-512
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Answer: A. MD5

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- A. Larger key size
- B. Smaller hash
- C. No hashing
- D. Plaintext

Answer: A. Larger key size

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- B. CA signature
- C. Hash only
- D. Encryption

Answer: B. CA signature

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- B. Authentication
- C. Compression
- D. Routing

Answer: B. Authentication

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- A. MD5 only
- B. SHA + HMAC
- C. Plaintext
- D. Encryption only

Answer: B. SHA + HMAC